DeepSeek-R1: Incentivizing Reasoning in LLMs via Reinforcement Learning

MS AI & MS DS — Deep Reinforcement Learning Project

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Paper Overview

DeepSeek-R1 introduces a purely **reinforcement learning-based** approach to develop chain-of-thought reasoning in large language models — without any supervised reasoning traces. The central finding is that when given only math questions and correct answers, an LLM trained with GRPO spontaneously learns to self-verify, reflect, and reason in structured multi-step chains. The authors call this emergent behaviour the **'aha moment'**.

Old Approach	This Paper	Key Result
Requires thousands of hand-labelled chain-of-thought reasoning examples — expensive, slow, limits creativity.	Uses only (question, answer) pairs. Model discovers how to reason via RL reward signals.	Model self-learns step-by-step thinking, self-correction, and beats o1 on AIME 2024 math benchmarks.

Key Contributions of the Paper

Paper Contributions

- **DeepSeek-R1-Zero:** First model to show reasoning can emerge from pure RL with no supervised fine-tuning
- **GRPO Algorithm:** Memory-efficient RL — replaces the critic model with group-relative reward statistics
- **Rule-Based Rewards:** Simple correctness + format rewards replace expensive reward models
- **Distillation:** Reasoning capabilities transferred to 1.5B–70B models, making it accessible at small scale
- **Open Source:** Full model weights, training code, and dataset released publicly

GRPO Algorithm — How It Works

Group Relative Policy Optimization (GRPO) is the core RL algorithm. For every training question, the model samples a group of G responses. Each response is scored. The model then updates itself to increase the probability of high-scoring responses and decrease low-scoring ones — no human in the loop, no separate critic/value network.

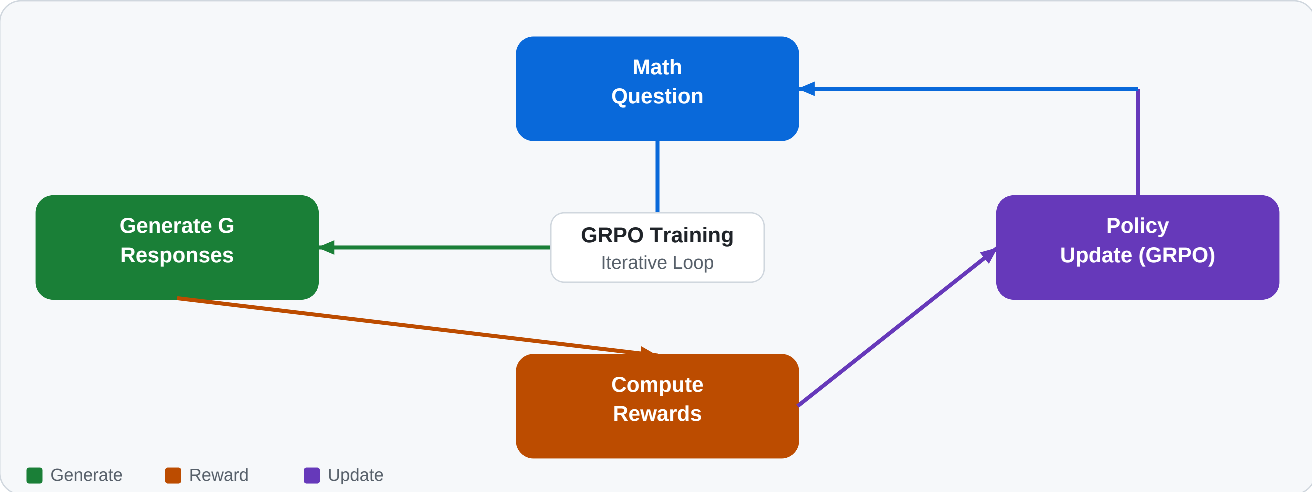

Figure 1: GRPO training cycle — model generates G outputs, scores them, and updates policy

Advantage Estimation (simplified)

Step	Formula	Meaning
1. Sample	o1, o2, ... oG ~ pi_theta(q)	Generate G responses from current policy
2. Score	r1, r2, ... rG = Reward(oi)	Score each response with reward function
3. Normalise	Ai = (ri - mean(r)) / std(r)	Advantage = how much better than group avg
4. Update	Maximise E[Ai * log pi_theta(oi|q)]	Push policy toward high-advantage outputs

Reward Function Design

The paper uses two simple, automatic reward functions that require no human annotation. The model must produce output in a specific format to receive any reward at all, which incentivises structured reasoning.

Correctness Reward

```
if answer == ground_truth:
reward = 1.0
else:
reward = 0.0
```

Format Reward

```
if <think>...</think> present
AND <answer>...</answer>:
reward = 0.5
else: reward = 0.0
```

Figure 2: Two complementary rewards — correctness (outcome) + format (process)

Expected Model Output Format

```
<think>
Let me solve this step by step...
```

```
[model reasons, self-corrects, verifies]
</think>
<answer> 42 </answer>
```

Reproduction Plan

We **will reproduce** the paper's core result — training a small LLM with GRPO to develop chain-of-thought reasoning using only question-answer pairs — using the **TinyZero** codebase (github.com/Jiayi-Pan/TinyZero) and the **Unsloth** library for memory-efficient training. This reproduction is the **primary deliverable** of the project and will be completed before any additional experiments are attempted.

Component	Choice	Reason
Base Model	Qwen2.5-1.5B	Smallest model that exhibits reasoning emergence
RL Algorithm	GRPO (Group Relative PO)	Paper's algorithm; no critic model needed
Training Data	GSM8K (grade school math)	Clean, verifiable Q&A; pairs with exact answers
Reward 1	Correctness (exact match)	Outcome-based, rule-based, no model needed
Reward 2	Format (think/answer tags)	Process-based; ensures structured reasoning
Evaluation	GSM8K test set accuracy	Standard benchmark used in the paper
Codebase	TinyZero + Unsloth	Minimal, open-source reproduction of R1-Zero

Reproduction Steps (Planned)

* **Step 1 — Baseline Evaluation:** Run Qwen2.5-1.5B (no RL) on GSM8K test set. Record accuracy as baseline.
* **Step 2 — Data Preparation:** Format GSM8K training set into (question, answer) pairs. No reasoning traces used.
* **Step 3 — Reward Implementation:** Code correctness_reward and format_reward functions matching paper specs.
* **Step 4 — GRPO Training:** Train for 500–1000 steps with G=8 generations per prompt, learning rate 1e-5.
* **Step 5 — Validate Aha Moment:** Monitor training logs — response length and reward should increase together.
* **Step 6 — Final Evaluation:** Evaluate trained model on GSM8K test set. Compare accuracy to Step 1 baseline.

Expected Outcomes from Reproduction

Metric	Base Model	After GRPO	Paper Reference
GSM8K Accuracy	~30-35%	~50-60%	Significant gain
Avg Response Length (tokens)	~50	~300+	Increases steadily
think Tag Usage	0%	~95%+	Emerges during training
Self-Correction Behaviour	None	Observed	Emergent property

Experiments — Subject to Project Timeline

Following successful reproduction, we **plan to conduct** up to five additional experiments to study what drives GRPO's success. **These experiments are timeline-dependent** — they will be executed in priority order (E1 → E5) based on remaining project time after the core reproduction is complete. High-priority experiments (E1, E2) are expected to be completed; medium and low-priority ones (E3–E5) will be attempted if time permits.

Timeline Note: Reproduction first — experiments second. E1 and E2 are strongly prioritised as they directly validate the paper's claims. E3 (generation count sweep) and E4 (model scale) require additional compute and will only be attempted if the project schedule allows. E5 (domain transfer) is a stretch goal.

	Experiment	What We Change	Research Question	Priority
E1	**Reward Ablation**	Remove format reward -> correctness only	Does format reward improve training signal?	High
E2	**GRPO vs Base LLM**	RL-trained vs untrained Qwen2.5-1.5B	Quantify reasoning improvement on GSM8K	High
E3	**Generation Count G**	Sweep G in {4, 8, 16} per prompt	More samples -> better policy estimate?	Medium
E4	**Model Scale**	Qwen2.5-1.5B vs Qwen2.5-3B	Does scale accelerate aha-moment?	Medium
E5	**Domain Transfer**	Train on math, eval on logic puzzles	Does GRPO generalise beyond math?	Low

E1 — Reward Ablation

High Priority

Train three variants: (A) both rewards, (B) correctness only, (C) format only. Measure GSM8K accuracy for each. **Hypothesis:** format reward provides crucial training signal during early steps when the model rarely gets answers correct. *Will be run if reproduction succeeds on schedule.*

E2 — RL vs No RL

High Priority

Compare the GRPO-trained model against the base Qwen2.5-1.5B on GSM8K. Show qualitative examples of structured reasoning vs. direct short guesses. **Hypothesis:** RL training yields 15–25% accuracy improvement. *Will be run alongside E1 as it reuses the same trained checkpoint.*

E3 — Generation Count G in {4, 8, 16}

Medium Priority

Vary the number of sampled responses per prompt. More samples give better advantage estimates but higher memory and compute cost. **Hypothesis:** G=8 gives the best accuracy-compute tradeoff. *Will be attempted if project timeline permits after E1 and E2.*

E4 — Model Scale (1.5B vs 3B)

Medium Priority

Run identical GRPO training on Qwen2.5-3B. Compare final accuracy and the step at which the aha moment first appears. **Hypothesis:** larger model reaches higher accuracy faster. *Requires additional GPU compute — conditional on resource and time availability.*

E5 — Domain Transfer

Stretch Goal

Train exclusively on math (GSM8K), then evaluate on a logic/puzzle dataset. **Hypothesis:** GRPO teaches general step-by-step reasoning beyond math pattern-matching. *Stretch goal — will only be attempted if all higher-priority work is complete.*

Project Summary: We will reproduce DeepSeek-R1's core GRPO training pipeline on Qwen2.5-1.5B using TinyZero/Unsloth and validate the emergence of chain-of-thought reasoning on GSM8K. Additional experiments (E1–E5) will be conducted in priority order based on the remaining project timeline — with E1 and E2 targeted as completed deliverables, and E3–E5 as best-effort stretch goals subject to available time and compute.